MagicLocomotion: Competition Solution For 2025 Multi-Terrain Humanoid Locomotion Challenge

Yue Chen[*1] Haoran Lu[†2] Chuye Hong[‡3] Ruihai Wu[*1] Hao Dong[*1]
[1]Peking University [2]Northwestern University [3]Tsinghua University

Abstract

The ICCV 2025 Multi-Terrain Humanoid Locomotion Challenge presents a rigorous testbed for developing robust locomotion controllers capable of navigating diverse, unstructured terrains. This report presents our solution that achieved 2nd place, demonstrating strong performance across three humanoid platforms (Unitree G1, Unitree H1-2, and Fourier N1). Our solution builds upon the official codebase and introduces two key innovations: (1) student policy distillation with carefully designed observation space compression, and (2) targeted fine-tuning for performance optimization. Through systematic ablations and architectural refinements, we demonstrate our method effectively learn locomotion skills across diverse terrain conditions. The code is https://github.com/yuechen0614/Magic-Locomotion

1. Introduction

Humanoid locomotion in unstructured environments remains one of the most challenging problems in embodied AI. Unlike wheeled or quadrupedal robots, humanoid platforms must maintain dynamic balance while executing high-dimensional control policies across terrains that vary drastically in geometry, friction, and compliance. The Multi-Terrain Humanoid Locomotion Challenge, hosted at ICCV 2025, systematically evaluates these capabilities through three complementary metrics: **robustness** (extended terrain sequences), **extreme performance** (maximum-difficulty scenarios), and **generalization** (mixed terrain combinations).

To address these issues, we introduce MagicLocomotion, a structured training framework that overcomes the limitations of fixed-terrain setups and extends to the multi-terrain humanoid domain. Our contributions are threefold:

- **Teacher Policy with Privileged Information**: We design a teacher policy that leverages comprehensive privileged observations, including terrain height maps and future terrain previews. The teacher demonstrates near-optimal performance across all terrain types, establishing an upper bound for student policy capabilities.
- **Progressive Teacher-Student Distillation**: Rather than direct behavior cloning, we employ a multi-phase distillation process that gradually transitions from teacher supervision to policy refinement. This approach prevents the student from overfitting to teacher actions while preserving critical locomotion skills.
- **Defect-Aware Fine-Tuning**: After distillation, we introduce targeted fine-tuning on specific terrain configurations where the performance gap between teacher and student remains large. This selective optimization allows us to address failure modes without sacrificing general robustness.

Through extensive experiments across three humanoid platforms and nine terrain types, we demonstrate that our framework achieves competitive performance with substantially improved sample efficiency.

2. Method

Our solution is built upon the official competition codebase Humanoid-Terrain-Bench. We decompose the problem into three sequential stages: training a teacher policy with full privileged information, distilling this knowledge into a student policy that operates on limited observations, and finally refining the student through targeted fine-tuning. This framework allows us to leverage rich environmental information during training while producing policies that can be deployed with only proprioceptive and terrain sensing.

2.1. Teacher Policy with Privileged Information

The teacher policy is trained using Proximal Policy Optimization with access to comprehensive environmental information beyond what is available at deployment time. The observation space includes standard proprioceptive measurements such as base angular velocity, orientation, joint positions and velocities, as well as privileged information including dense terrain height maps, friction coefficients at contact points, and terrain type encodings. Additionally, the teacher receives contact force measurements and future terrain previews that enable anticipatory control strategies.

The policy network consists of a three-layer encoder that processes the observation vector through hidden dimensions of 512, 256, and 128 with ELU activations. This shared representation feeds into separate policy and value heads. The policy head outputs mean joint position targets, while the value head provides state value estimates for advantage calculation.

We employ a standard reward function that combines velocity tracking, orientation maintenance, base height stabilization, action smoothness penalties, and energy efficiency terms. Importantly, we introduce the curriculum during training by first establishing basic locomotion on flat terrain before gradually exposing the policy to increasingly challenging terrain configurations. This staged curriculum prevents premature convergence to poor local optima and ensures the teacher develops robust behaviors across the full terrain distribution. The teacher is trained for 30 million environment steps across 4096 parallel simulation instances.

2.2. Progressive Teacher-Student Distillation

We formulate distillation as a combined objective that interpolates between behavioral cloning and independent reinforcement learning. The loss function is defined as:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{BC}} + (1 - \alpha)\mathcal{L}_{\text{PPO}}, \tag{1}$$

where $\mathcal{L}_{\text{BC}}$ measures the KL divergence between teacher and student action distributions, and $\mathcal{L}_{\text{PPO}}$ denotes the standard PPO objective for the student. The mixing coefficient α determines the relative emphasis on imitation versus independent learning.

Rather than using a fixed α, we employ progressive scheduling that initializes at $\alpha = 0.8$ and decays linearly to $\alpha = 0.2$ over the course of distillation. This schedule enables the student to initially rely heavily on teacher supervision, while gradually transitioning to independent policy improvement as its own value function becomes better calibrated. The distillation phase is executed for 10 million steps with a reduced learning rate of 10^{-4} to prevent catastrophic forgetting of the teacher's demonstrated behaviors.

Critically, we perform online distillation by collecting fresh teacher rollouts throughout the entire process, instead of distilling from a fixed offline dataset. This ensures the student observes teacher behavior across the full state distribution—including error recovery and perturbation handling, which would be underrepresented in demonstration data collected only from successful episodes.

2.3. Defect-Aware Fine-Tuning

After distillation, we analyze the performance gap between teacher and student across different terrain configurations. While the student achieves near-teacher performance on most scenarios, systematic failures emerge on particularly challenging terrain. These failure modes suggest that certain skills requiring precise privileged information were not fully transferred during distillation.

To address these gaps, we perform targeted fine-tuning on a curated set of hard scenarios identified through validation set analysis. For each difficult terrain type, we construct a focused sub-curriculum that gradually increases difficulty within that specific domain. Fine-tuning uses conservative hyperparameters with learning rate 5e-5 and incorporates experience replay from successful Stage 2 trajectories to prevent performance degradation on already-solved tasks. We also apply value function regularization that penalizes large deviations from the post-distillation value estimates.

3. Environment Analysis

This section analyzes the challenge environment, terrain distribution, and task complexity to provide context for our design decisions.

3.1. Challenge Setup

The Multi-Terrain Humanoid Locomotion Challenge evaluates policies across three humanoid robots, nine terrain types, and three evaluation tracks. The robot platforms include Unitree H1-2 (19 DoF), Unitree G1 (29 DoF), and Fourier N1 (23 DoF), each with distinct kinematic configurations. The terrain suite covers flat ground, stairs, slopes, gaps, hurdles, bridges, uneven ground, wave terrain, and parkour courses—encompassing both basic locomotion and complex multi-skill scenarios. The three evaluation tracks target key performance dimensions: Robustness (extended terrain sequences, e.g., 20+ consecutive stairs versus 5–10 in training), Extreme (maximum difficulty configurations, e.g., 0.25m stair height versus the nominal 0.15m), and Generalization (unseen mixed terrain combinations). Policies are scored using a weighted metric:

$$\begin{aligned} \text{Score} &= 0.9 \times \text{CompletionRate} \\ &\quad + 0.1 \times (1 - \text{EfficiencyScore}) \end{aligned} \tag{2}$$

where CompletionRate measures distance coverage and EfficiencyScore reflects time-to-completion.

Terrain Characteristics We analyzed the difficulty profile of each terrain type using training data, with key findings summarized as follows. Flat ground serves as the baseline with an average episode length of 500 steps and a 99.8% success rate for the teacher policy. Stairs—accounting for 30% of training episodes—require precise foot placement, achieving a 94.2% teacher success rate over 350-step episodes. Slopes (400 steps, 96.5% success) demand balance under gravity, while gaps (320 steps, 88.7% success) test jumping and dynamic balance capabilities, with success rates varying widely (65%–98%) based on gap width. Hurdles (340 steps, 90.1% success) require adequate leg clearance, and uneven ground (450 steps, 91.8% success) challenges robots with irregular footing. Wave terrain (380 steps, 93.4% success) involves oscillatory balance, while bridges (280 steps, 82.3% success) and parkour courses (420 steps, 85.6% success) present the highest fundamental difficulty—marked by narrow support areas for bridges and multi-skill composition for parkour—resulting in the lowest teacher success rates.

3.2. Robot Platform Differences

Each humanoid platform exhibits unique strengths and limitations that influence terrain adaptability. Unitree H1-2 is mechanically robust with well-documented kinematics but suffers from limited hip rotation and a tendency to overshoot on stairs, performing best on flat ground, slopes, and uneven terrain. Unitree G1's high DoF count enables complex maneuvers and excellent balance, though it has a slower maximum velocity and sensitivity to payload changes—excelling in parkour, bridges, and gaps. Fourier N1 strikes an intermediate complexity balance with strong generalization, but is less stable on extreme slopes, with optimal performance on stairs, hurdles, and wave terrain. Our multi-robot training framework accounts for these differences through platform-specific reward weights and dedicated validation sets.

3.3. Evaluation Protocol

The competition employs a standardized evaluation server to ensure fair comparison and prevent overfitting to local simulation parameters. Policies are compiled to TorchScript format, loaded by the server, and executed in the Isaac Gym simulator. Episodes initiate with env.reset(), with the policy receiving observations via a REST API and returning actions in real time. Episodes terminate upon robot fall (base height below a predefined threshold) or timeout (maximum 1000 steps). Scores are computed per episode and aggregated across all test scenarios, ensuring consistent assessment of policy performance across platforms and terrains.

4. Conclusion

We present a novel learning framework enhanced with progressive distillation and targeted fine-tuning for solving complex multi-terrain humanoid locomotion tasks. By leveraging teacher-student architecture with privileged terrain observations, transitioning gradually from supervision to independent learning, and addressing residual failure modes through selective optimization, our approach achieves robust, generalizable performance across three humanoid platforms and nine terrain types. We believe this paradigm provides a strong foundation for tackling the Multi-Terrain Humanoid Locomotion Challenge and contributes valuable insights for bridging simulation and reality in embodied AI.

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